

JEDEC STANDARD

Preconditioning of Nonhermetic Surface Mount Devices Prior to Reliability Testing

JESD22-A113I

(Revision of JESD22-A113H, November 2016)

APRIL 2020

JEDEC SOLID STATE TECHNOLOGY ASSOCIATION



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Published by
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TEST METHOD A113I**PRECONDITIONING OF NONHERMETIC SURFACE MOUNT DEVICES
PRIOR TO RELIABILITY TESTING**

Foreword

This document provides an industry standard test method for preconditioning packaged devices that is representative of a typical industry multiple solder reflow operation.

Introduction

The typical use of surface mount devices (SMD) involves subjecting the SMDs to elevated temperatures during board assembly, which, by itself or combined with moisture in the package can induce internal package damage that could be a reliability concern. Preconditioning of SMD packages is used to simulate the effects of board assembly prior to reliability testing. This allows reliability testing at the packaged device level on as shippable products with a board assembly simulation. During preconditioning, test samples are subjected to temperature cycling (optional), dry bake, moisture soaking, solder reflow simulation, flux, rinse, dry, and electrical test before reliability testing.

This test method references the reflow profiles stated in J-STD-020. If a packaged device is not able to withstand the full thermal profile as stated in J-STD-020, J-STD-075 should be used to evaluate and classify process sensitivities.

Diamond Foundry

TEST METHOD A113I**PRECONDITIONING OF NONHERMETIC SURFACE MOUNT DEVICES
PRIOR TO RELIABILITY TESTING**

(From JEDEC Board ballot JCB-20-10, formulated under the cognizance of the JC-14.1 Subcommittee on Reliability Test Methods for Packaged Devices.)

1 Scope

This Test Method establishes an industry standard preconditioning flow for nonhermetic solid state SMDs that is representative of a typical industry multiple solder reflow operation. These SMDs should be subjected to the appropriate preconditioning sequence of this document by the manufacturer prior to being submitted to specific in-house reliability testing (qualification and reliability monitoring) to evaluate long term reliability (which might be impacted by solder reflow).

Devices that could be surface mounted, but are designed/intended to only be socketed, are out of scope of this test method.

NOTE For good correlation of results between moisture/reflow-induced stress sensitivity testing (per J-STD-020 and JESD22-A113) and actual reflow conditions used, identical temperature measurements by both the SMD manufacturer and the board assembler are necessary. Therefore, it is recommended that the package temperature at the top center of the package be determined during assembly board reflow profile setup, to ensure that it does not exceed the evaluation temperature based on package thickness and volume as stated in J-STD-020.

2 Normative reference

IPC/JEDEC J-STD-020, *Moisture/Reflow Sensitivity Classification for Nonhermetic Solid State Surface Mount Devices*

IPC/JEDEC J-STD-033, *Handling, Packing, Shipping and Use of Moisture/Reflow Sensitive Surface Mount Devices*

ECIA/IPC/JEDEC J-STD-075, *Classification of Passive and Solid State Devices for Assembly Processes*

JESD22-B101, *External Visual*

JESD22-A104, *Temperature Cycling*

JESD625, *Requirements for Handling Electrostatic Discharge Sensitive (ESD) Devices*

JESD47, *Stress-Test-Driven Qualification of Integrated Circuits*

JESD94, *Application Specific Qualification Using Knowledge Based Test Methodology*

3 Definitions

For the purposes of this standard, the following terms and definitions apply:

crack: A separation within a bulk material.

delamination: An interfacial separation between two materials that were intended to be bonded.

4 Apparatus

This test method requires use of the following equipment.

4.1 Temperature and humidity chamber

Moisture chamber(s) capable of operating at 85 °C/85% RH, 85 °C/60% RH, and 30 °C/60% RH. Within the chamber working area, temperature tolerance must be ± 2 °C and the RH tolerance must be $\pm 3\%$ RH. A chamber with 60 °C/60% RH capability is optional for accelerated soak conditions (See J-STD-020).

4.2 Solder reflow equipment

- a) (Preferred) - Full Convection reflow system capable of maintaining the reflow profiles required by this standard.
- b) Infrared (IR)/Convection solder reflow equipment capable of maintaining the reflow profiles required by this standard. It is required that this equipment use IR to heat only the air and not directly impinge upon the SMD packages under test.

NOTE The moisture/reflow and process sensitivity classification test results are dependent upon the package body temperature (rather than the reflow carrier and/or package terminal temperature).

4.3 Optical microscope

Optical Microscope (40x for external visual exam).

4.4 Electrical test equipment

Electrical test equipment capable of performing room temperature dc and functional tests.

4.5 Bake oven

Bake oven capable of operating at 125 $\pm 5/0$ °C.

4.6 Temperature Cycle Chamber

Temperature Cycle Chamber capable of operating, as a minimum, over the range of -40 °C (or lower) to 60 °C (or higher) per JESD22-A104. This equipment is only required if optional Step 5.3 is used.

4.7 Acoustic Microscope

Acoustic microscope capable of detecting delamination on any critical interface as defined in Annex B1. Either C-Mode or Through Transmission Mode may be used.

NOTE Refer to IPC/JEDEC J-STD-035 for operation of the acoustic microscope.

5 Test procedure

Using similar SMDs, it is recommended that the moisture sensitivity level (MSL), per J-STD-020, be determined before starting the preconditioning sequence to establish which moisture soak condition is appropriate, i.e., likely to pass. If the MSL level is not known then other relevant moisture evaluation data may be consulted, or an arbitrary selection may be made. Multiple moisture soak conditions can also be run to determine a passing level. However, the soak condition used must be consistent with the floor life information in J-STD-020.

Reflow requirements are provided for both Pb-free and legacy SnPb conditions and should be used based on the intended end use of the SMD. The same package may have different MSL levels depending on whether the SnPb or Pb-free reflow is used.

Some SMDs with a unique construction or materials may have limitations independent of moisture exposure that are strictly due to the thermal profile stated in J-STD-020 and may not be able to fully meet one or several of the temperature and/or duration requirements of the reflow profile. Some SMDs may have other limitations due to other assembly processes, such as exposure to chemicals. For those SMDs with thermal and/or other process limitations follow the process sensitivity classification stated in J-STD-075 to identify the appropriate process sensitivity level (PSL) rating.

For example, if a SMD cannot meet its required peak temperature based on its volume and thickness, then the peak temperature of the profile used for the preconditioning must be changed based on the value stated by its PSL classification. Thus, if the SMD should be subjected to 260 °C, but its PSL rating is R6, the peak temperature used for preconditioning shall be 250 °C.

Similar to MSL, if the PSL level is not yet determined, relevant evaluation/engineering data may be used to generate the initial profile parameters. If the SMD passes the subsequent qualification testing, then the PSL rating will be based on the chosen parameters used for preconditioning; however, if there are fails that are deemed to be due to the thermal exposure of the preconditioning flow, requalification is required with reduced profile conditions that allow the SMD to pass the qualification requirements.

At all times the test parts should be handled using proper ESD procedures in accordance with JESD625.

Refer to Annex A for the typical test flow.

NOTE If the preconditioning sequence is being performed by the semiconductor manufacturer, steps 5.1, 5.2, 5.4, 5.10 and 5.11 are optional since they are the supplier's risks. If the preconditioning sequence is being performed by the user, steps 5.7 through 5.9 are optional.

5.1 Initial electrical test

Perform electrical and/or functional test to verify that the SMDs meet the room temperature data sheet specification. Replace any test samples that fail to meet this requirement.

5.2 Visual inspection

Perform an external visual examination under 40X optical magnification (per JESD22-B101 or agreed to alternative) to ensure that no samples with external cracks or other damage are used in this test method. If mechanical rejects are found, corrective action must be implemented in the manufacturing process and a new sample must be drawn from product that has been processed with the corrective action. If there is no plan to implement corrective actions, all mechanical rejects must be kept and put through this test method and subsequent reliability stress tests.

Optionally, perform acoustic microscopy scan to assess initial adhesion of key interfaces and then track any progression due to preconditioning and stress testing. This may be performed on a sample of the devices which then should not be put on reliability stress tests. If performed, devices should be serialized so “pre-reflow” and “post-reflow” images can be compared.

5.3 Temperature cycling

Perform five (5) cycles of temperature cycle from -40 °C (or lower) to 60 °C (or higher) to simulate shipping conditions. Acceptable alternative test conditions and temperature tolerances are A through I, L through N, and T as defined in Table 1 of JESD22-A104, Temperature Cycling. This step is optional based on product requirements.

5.4 Bake out

Bake the samples for 24 hours minimum at 125 +5/-0 °C. This step is intended to remove all moisture from the package so that it will be “dry.”

NOTE 1 This time/temperature may be modified if desorption data on the particular SMD being preconditioned shows that a different condition is required to obtain a “dry” package. Refer to J-STD-020 for procedures on running absorption and desorption curves.

NOTE 2 If the SMD cannot be subjected to a bake at 125 °C for 24 hours or longer, then use a lower temperature condition with a longer duration to ensure that all moisture is removed. See J-STD-033 for alternative bake conditions and durations that are equivalent.

5.5 Moisture Soak

Place samples in a clean, dry, shallow container so that the package bodies do not touch or overlap each other. Submit each sample to the appropriate moisture soak requirements shown in J-STD-020. The moisture soak should be initiated within 2 hours of bake.

NOTE The moisture soak is optional for SMDs (e.g., Flip Chip / bumped die devices, etc.) where moisture absorption data is available showing the particular SMD being preconditioned either does not absorb moisture or desorption is so fast that the preheat portion of the reflow cycle will remove all moisture. Alternatively, the moisture soak can be abbreviated to the point in time where saturation occurs based on available absorption data. Refer to J-STD-020 for procedures on running absorption and desorption curves.

5.6 Reflow

Not sooner than 15 minutes and not longer than 4 hours after removal from the temperature/humidity chamber, subject the sample to 3 cycles (See NOTE 1) of the appropriate reflow conditions. The reflow conditions are defined in J-STD-020. If the timing between removal from the temperature/humidity chamber and initial reflow cannot be met, then the parts must be rebaked and resoaked according to 5.4 and 5.5.

NOTE 1 The 3 reflow cycles represent the following:

- Cycle 1 - the first pass of a Double-Sided, Double-Pass (DSDP) assembly reflow process.
- Cycle 2 - the second pass of a DSDP assembly reflow process.
- Cycle 3 - rework of a near neighbor on the assembly where the SMD being classified experiences reflow-like temperatures.

NOTE 2 For SMDs classified per J-STD-075 that cannot be subjected to 3 reflows, only perform the number of reflows for which the SMD is classified.

NOTE 3 If the reflow cycle regimen is not representative for the SMD being classified, refer to JESD94 for application specific qualification guidance. The following is an example of an application that would require an addition reflow cycle. SMDs that are placed directly next to a CPU in an LGA socket or on the underside of the PWB from an LGA socket (e.g., power management devices) may see 4 reflow cycles due to rework of the LGA socket. In that case, evaluation of 4 reflow cycles may be necessary to account for the total thermal exposure. Reflow cycle 3 would represent the removal of a damaged LGA socket and reflow cycle 4 would simulate the attaching of the replacement socket.

The sample parts shall be cooled sufficiently (preferably back to room temperature) between reflow cycles so that the reflow temperatures/times of the samples are not affected on the subsequent reflow cycles.

Reflow practices shall be sufficient to ensure that all sample parts, in each reflow cycle, will meet the appropriate reflow profile requirements of J-STD-020.

SMDs intended for use in a “Pb-free” assembly process shall be evaluated using the “Pb-free” reflow temperature whether or not the SMD is Pb-free.

If parts are reflowed in other than the normal assembly reflow orientation (i.e. live bug/dead bug) the damage response should be correlated.

The reflow oven should be loaded with the same configuration or be verified to have an equivalent thermal load when running preconditioning as was used to develop the reflow profile.

The reflow profiles in J-STD-020 are only for classification and preconditioning and are not meant to specify board assembly profiles. Actual board assembly profiles should be developed based on specific process needs and board designs and should not exceed the parameters in J-STD-020.

5.6.1 Solder attachment after reflow

If reliability testing is to be performed in a second level configuration, one of the reflow cycles may be used to attach the sample to the test board. If the board assembly is to be performed at a later time than the samples, at the user discretion, can be baked and vacuum sealed until such time that it is solder attached to the test board or facsimile. The profile used to attach the sample to the test board may be optimized for the soldering operation, but the other two passes must meet the profile requirements of J-STD-020.

NOTE Bumped die devices are commonly soldered to test boards, due to the potential of inflicting handling damage on the loose devices while performing the preconditioning, qualification stresses, and functional testing. If the third reflow pass is used to solder the bumped die devices to boards, care must be taken to minimize damage to the bumped die. If the damage is severe, it could cause devices to fail electrically during the next functional test; if the damage is less severe, it could weaken devices that then fail during reliability stress testing. It may be difficult to determine the root cause for fails if handling damage has occurred.

Since the board attachment replicates a real-life process with flux application, reflow and cleaning, steps 5.7, 5.8, and 5.9 are no longer necessary or mandated prior to submission to reliability stress testing. Flux type shall be documented per clause 7.

5.7 Flux application

After the reflow solder cycles are completed, allow the samples to cool at room ambient for 15 minutes minimum. Samples shall then be dipped (full body immersion) in an activated water soluble flux at room ambient temperature for 10 seconds minimum.

LGA packages that are soldered attached (as opposed to LGAs that are socketed) do require flux application.

Flux application is only performed on samples for high humidity stresses (e.g., HAST, autoclave, Temperature - Humidity Bias, etc.).

5.7.1 Criteria for when flux application is not performed

Ball Grid Array (BGA) packages with a nominal standoff of greater than 0.2 mm and all Column Grid Array (CGA) packages do not require flux dipping because during their typical board assembly processes liquid fluxes do not come in contact with these package types. Steps 5.8 and 5.9 are not required when flux dipping is omitted for these package types.

Flux application, step 5.8, and step 5.9 are to be omitted for any SMD that has a flux or washing limitation as stated as part of its PSL rating (per J-STD-075).

Flux application is not required for any package type if it can be shown that there is not a path for flux, rinse water, and other contaminants to enter a delaminated interface or package crack (as received or caused by solder reflow conditions) and contact a critical feature within the device that could cause an immediate failure or create a reliability concern. See Annex B for the procedure to determine if flux application may be omitted. As an option, this evaluation can be performed on samples that are being used for MSL determination per J-STD-020, if the preconditioning samples are similar (See Annex B).

5.8 Cleaning

Clean samples using multiple agitated deionized water rinses. No waiting time is required between flux application and cleaning. Ensure that all flux residuals are completely removed.

5.9 Drying

Samples should be dried at room ambient prior to submission to reliability testing.

5.10 Post Reflow Visual Inspection

A post reflow visual inspection is recommended to identify any package cracks that may have been created due to the preconditioning process and ensure that the devices still meet the physical outline requirements.

If a post preconditioning acoustic microscopy scan is to be performed, it can be performed at this time.

5.11 Final electrical test

Submit the samples to electrical and/or functional testing per the room temperature data sheet specification. For the semiconductor manufacturer, this step is optional and may be omitted since it is a supplier's risk. Any valid failures found at this point due to the preconditioning sequence indicate that the SMD may have been classified in the wrong level (MSL and/or PSL) or something is substandard with the test samples. Failure analysis should be conducted. If appropriate, this SMD type should be reevaluated to determine the correct moisture and/or process sensitivity level. This would require resubmitting a sample to a new preconditioning sequence that has been appropriately modified prior to reliability testing per 6.

6 Applicable reliability tests

SMDs should be subjected to the appropriate preconditioning sequence of this document prior to being submitted to reliability tests per JESD47 or the semiconductor manufacturer's in-house reliability procedures.

7 Summary

The following details shall be specified in the applicable procurement document.

- a) Reflow profile used, either state revision of J-STD-020 or profile conditions per J-STD-075 (including PSL rating).
- b) Soak conditions if other than the standard conditions/durations stated in J-STD-020.
- c) Number of reflow cycles if other than three.
- d) Whether flux application is required, and if required, type of flux if other than 5.7.
- e) Reliability tests if other than 6.
- f) Test conditions and duration of reliability tests in 6.
- g) Electrical test description, including test temperature(s).
- h) Criteria and conditions for visual inspection and (if performed) acoustic microscopy (AM) scans if other than 5.2.

Annex A (normative) Typical Preconditioning Sequence Flow

Step	Item	Details	Section
1	Initial Electrical Test	- Replace any failing samples - Optional for testing by Supplier	5.1
2	Visual Inspection	- Replace any failing samples only if corrective actions have been implemented to remove the defect - Optional for testing by Supplier - Optional Acoustic Microscopy (AM) scan	5.2
3	Temperature Cycling	- 5 cycles -40 °C to 60 °C - Optional shipping simulation based on product requirements	5.3
4	Bake	- 24 hours at 125 °C - Optional for testing by Supplier	5.4
5	Moisture Soak	- Soak time and conditions per J-STD-020 based on the MSL level	5.5
6	Reflow	- 3 reflow cycles using profiles per J-STD-020, document revision of J-STD-020 used - SnPb or Pb-free profile based on SMD end use process	5.6
7	Flux Application	- 10 seconds full immersion in activated water-soluble flux - Optional for testing by User or second level configuration - Not required for CGA and high standoff BGA packages - See Annex B for criteria to determine if required - Perform analysis (visual, AM scan, dye penetration, etc.,) to determine if flux application required. Can be performed as part of MSL determination per J-STD-020.	5.7
8	Cleaning	- DI water rinse - Remove all flux residual - Optional for testing by User or second level configuration - Not required for CGA and high standoff BGA packages or if flux application is not required per Annex B	5.8
9	Drying	- Room ambient drying - Optional for testing by User or second level configuration - Not required for CGA and high standoff BGA packages or if flux application is not required per Annex B	5.9
10	Post Reflow Visual Inspection	- Optional, but recommended inspection to identify any reflow induced defects that do not meet product specification or workmanship requirements - Optional AM scan	5.10
11	Final Electrical Test	- If all samples pass, then ready for Reliability Testing - If valid failures are found, then either samples may need to be tested to a lower MSL or PSL classification level or something is substandard with the samples and root cause analysis needs to be performed - Optional for testing by Supplier, but recommended	5.11

Annex B (normative) Process for determining if flux application must be performed

B.1 Criteria

Flux application can be omitted if after preconditioning there is irrefutable evidence on a statistically significant sample from multiple assembly lots of the device that no delamination nor any other internal defect or damage exists that would allow flux, water, or any other contaminant to penetrate from the outside of the package to:

1. the point of the connection of any wire bond or flip chip bump to the lead frame or substrate;
2. to any surface of the die;
3. to any feature within the package susceptible to corrosion that could affect the function of the device (e.g., open, leakage/short, parametric, etc.).

B.2 Process

This evaluation can be performed on the same samples used for determining MSL classification as part of J-STD-020, but beware that the criteria in B.1 is different from the criteria in J-STD-020.

After having performed 3 passes of reflow simulation:

At random, choose a minimum of 22 devices split equally between all lots. A minimum of two nonconsecutive assembly lots is required.

Initial visual inspection, (40x magnification) for cracks, separation of mold compound from leadframe/substrate

- Per clause 6.1 of J-STD-020, no external cracks are allowed after reflow, if a crack is seen the MSL classification must be rerun at a different MSL level.
- If any external separation is observed between the encapsulation material and leadframe/substrate (delamination) consider modifying the J-STD-020 MSL level and restarting preconditioning (new samples, different moisture soak condition, perform 3 reflow passes).

Evaluate the package construction to determine if an acoustic microscope (AM) scan would be able to identify all external paths to any delaminated interface within the package; thus, determining whether the dye penetration test must be performed or, as an option, the water penetration test can be used. See Annex C for criteria for determining if AM can be used to verify all the criteria in B.1.

Perform acoustic microscopy scans on all samples and identify all samples with delamination.

If AM can be used to verify all criteria in B.1, then:

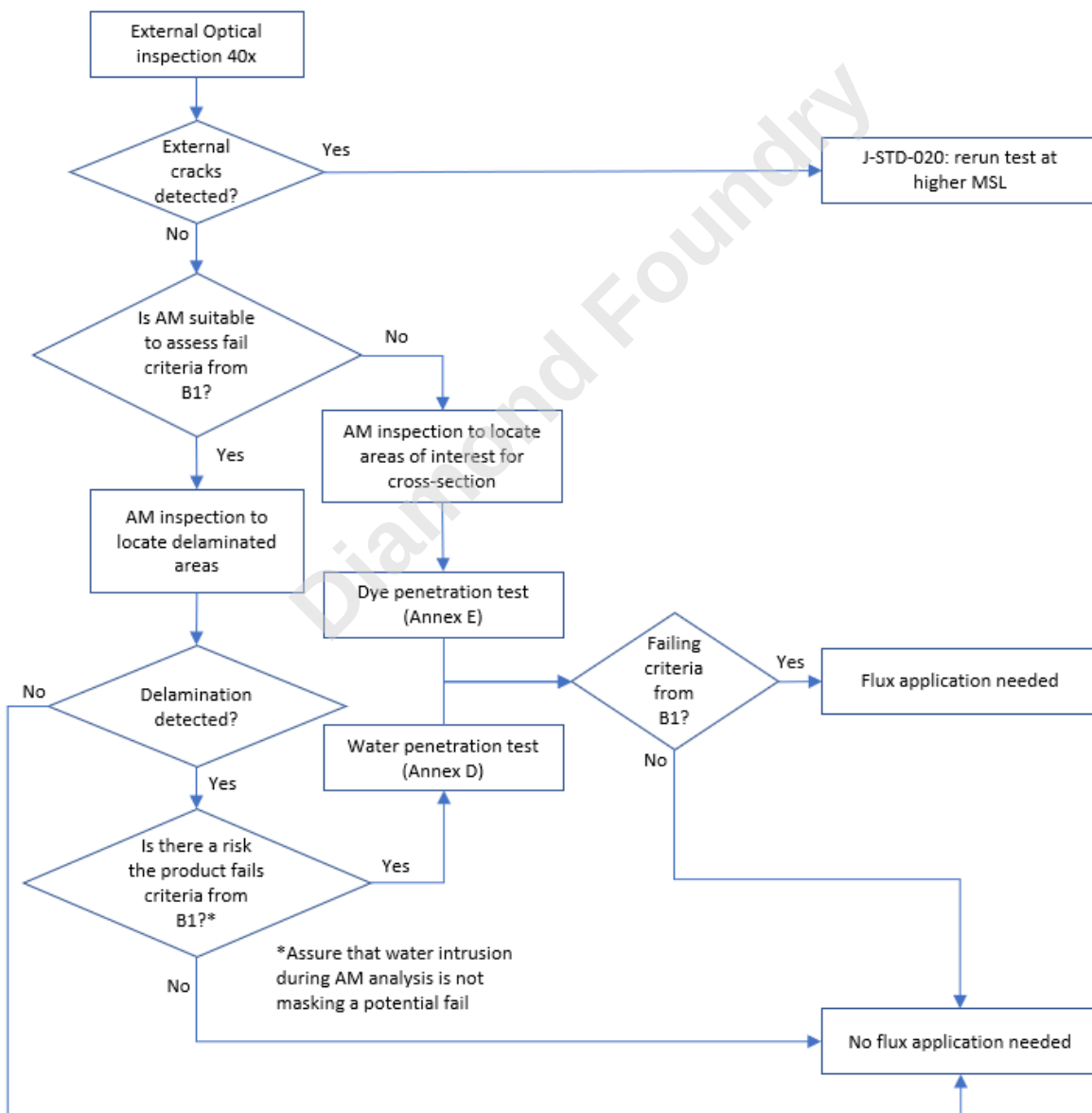
- If delamination is observed per the criteria in B.1, then perform water penetration per Annex D to determine if flux application is required. Water penetration is to be performed only on those samples showing delamination.
- If no delamination is observed, or delamination is observed but does not exceed the criteria stated in items 1, 2, and 3 in B.1, then flux application does not need to be performed.

B.2 Process (cont'd)

If AM cannot be used to verify all of the criteria in B.1, then perform dye penetration test as follows:

- Choose a minimum of 3 devices that exhibit the most severe delamination/damage per the AM scans and are most likely to have the delamination extend to an external surface.
- If the location of the delamination varies within the samples, which may suggest multiple mechanisms, different mechanisms, more samples may be required to assess the level of risk
- If no delamination is observed choose 5 devices at random.
- Perform dye penetration test according to Annex E to determine if there are any paths for contaminants to enter the package.
- See Annex F for hints on deciding where to perform cross sections after dye penetration.

Figure B.1 — Diagram of above process flow



Annex C (normative) Criteria for determining if AM can be used to verify all criteria in B.1

As referenced in B.2, this annex provides guidance on whether the device to be preconditioned can be evaluated using the water penetration test or must be subjected to the dye penetration test. As stated in B.1, flux application can be omitted if after preconditioning there is irrefutable evidence on a statistically significant sample from multiple assembly lots of the device that no delamination nor any other internal defect or damage exists that would allow flux, water, or any other contaminant to penetrate from the outside of the package to:

1. the point of the connection of any wire bond or flip chip bump to the lead frame or substrate;
2. to any surface of the die;
3. to any feature within the package susceptible to corrosion that could affect the function of the device (e.g., open, leakage/short, parametric, etc.).

Therefore, if acoustic microscopy (AM) is to be the sole tool used to determine if a path exists, AM must be able to accurately scan all interfaces that could become delaminated.

Annex G provides information on the capabilities of acoustic microscopy (AM) and how package features may affect the AM response. AM is not able to detect any delaminated interface that is perpendicular with the top surface of the package. If a package has any leadframe features that are exposed on the bottom of the package, AM is not able to determine if any of the vertical interfaces on the leadframe are delaminated (See Example 1). Similarly, AM requires the top surface of package to be flat and perpendicular to the direction of the AM signal. Thus, depending on the flatness of the top of a package or the shape of the outside top edge of the package (curvature, bevel, etc.), AM may not be able to obtain accurate information underneath any non-perpendicular surface. Example 2 describes how significant curvature of the outside edge of a TSSOP package affects the AM signal.

Annex C (normative) Criteria for determining if AM can be used to verify all criteria in B.1 (cont'd)

EXAMPLE 1 Flipchip QFN with vertical interfaces

For this example, a device in a flipchip QFN package was submitted for AM and the reflection mode scan (through the top surface) only detected a possible issue on one terminal (See Figure C.1). The device was turned over and an AM scan was performed through the bottom surface (See Figure C.2). The AM scan still detected an issue with the same terminal, but it was difficult to discern if this or other areas along the edge of other terminals were ~~real~~ truly delaminated or were just scattered signals due to edge effects. Figure C.3 is an optical image of the bottom of the device, with the red line showing where the package was cross-sectioned for the dye penetration analysis.

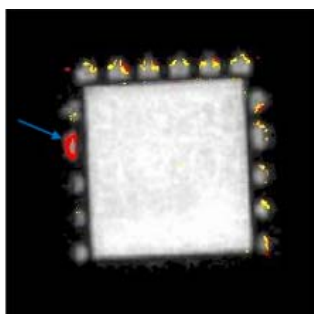


Figure C.1

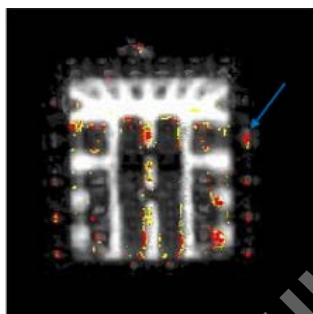


Figure C.2

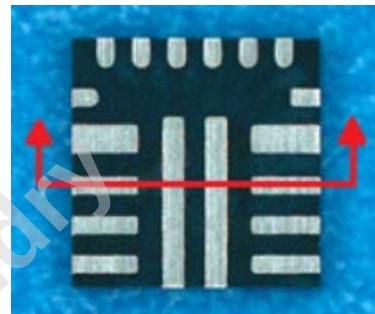


Figure C.3

Figure C.4 and Figure C.5 are optical images of the cross section that was performed on this device after dye penetration. In Figure C.6 the cross section of the device after dye penetration shows the penetrating material, which in this case fluoresces green, going up the side of the leadframe and the flip chip bump and contacting the die. The polyimide coating on the die, like other organic materials, may fluoresce and in this case fluoresces yellow, as seen in Figure C.6.

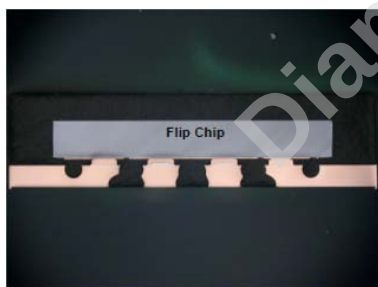


Figure C.4



Figure C.5

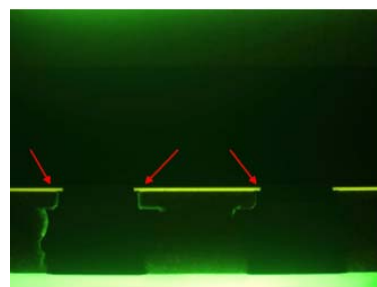


Figure C.6

In conclusion, AM was not able to accurately scan all surface breaking interfaces for this flipchip QFN package and thus dye penetration testing would be required; and since the dye penetration results show a path from the exterior of the package to the surface of the die, flux application would be required for this device.

EXAMPLE 2 TSSOP with bevel and rounded edges

For this example, the device consists of a smaller die on top of a larger die in a TSSOP package. Per Figure C.7, this package does not have any surface breaking features on the bottom of the package, thus there was a possibility of being able to use AM. An AM reflection mode scan was performed through the top surface of the device (Figure C.8) and there appears to be some delamination on the top surface of the leadframe, thus the question is whether AM is able to determine if a path exists from the outside, along the lead finger, and reaching the (stitch) wirebond on any of the lead fingers. Figure C.7 shows that this package has a significant bevel and rounded edges and as shown in G.3 in Annex G nearly 2/3 of the lead finger cannot be accurately scanned using AM, specifically the portion of the lead finger that exits the package; thus, AM cannot determine if there is an external path to any of the lead fingers with delaminated surfaces.

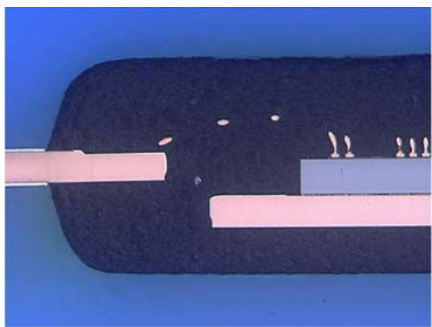


Figure C.7

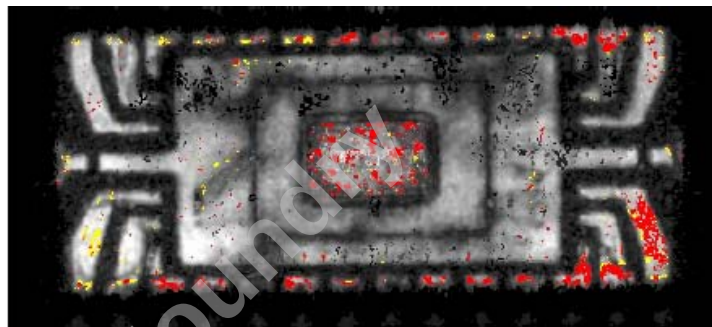


Figure C.8

Figure C.9 shows where the package was cross-sectioned after dye penetration. The fluorescing images in Figure C.10 show that the entire top surface of the lead finger is delaminated and Figure C.11, which is a higher magnification of C.10, shows that the wirebond itself is delaminated as dye is shown around the wire.

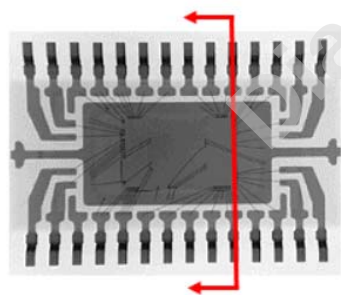


Figure C.9

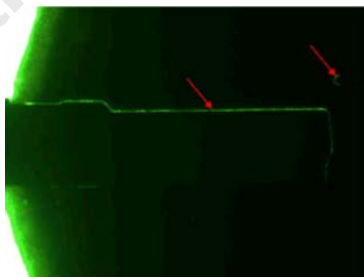


Figure C.10

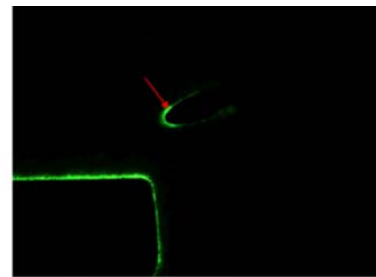


Figure C.11

In conclusion, AM was able to scan interior portions of all of the surface breaking interfaces for this TSSOP package, but it could not scan where the features exited the package. As delamination was detected on the lead fingers, dye penetration testing was required; and since the dye penetration results (which in this image fluoresces green) showed a path from the exterior of the package to where the wire (stitch) bond was on the lead finger, flux application was required for this device.

EXAMPLE 3 LQFP with rounded edges

This example is similar to Example 2, but in this case, it is an LQFP with rounded edges. Figure C.12 shows a cross section what was performed on this device after dye penetration testing. Note that for this device the wire bonds are much further in within the package and not near the outside edge as was the case for the TSSOP. Also note that this package does not have any surface breaking features on the bottom side, though it does have a unique die paddle design.



Figure C.12

Figure C.13 is an x-ray image of the device and Figure C.14 is an AM scan through the top surface of the package. The four white areas outlined in red is the polyimide tape that is used to protect the lead fingers during manufacturing. As the tape material is of a lower density than the leadframe and the molding compound the AM scan detects it, but a quick review of the acoustic signal will show that the tape is not delaminated.

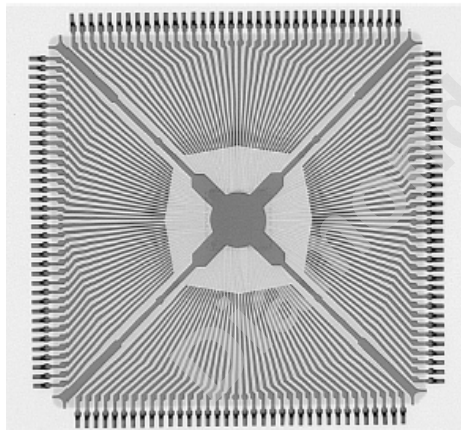


Figure C.13

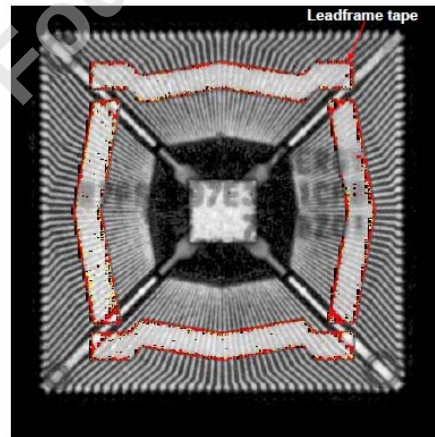


Figure C.14

In conclusion, AM was able to scan interior portions of all of the surface breaking interfaces for this LQFP package, but it could not scan where the features exited the package. However, in this case no delamination was detected within the package, so not only could water penetration testing be used in lieu of dye penetration for this package, flux application could be omitted for this device.

Annex D (normative) Water penetration test method

Refer to Annex B to determine for which samples this test method is appropriate.

The apparatus necessary to perform the water penetration test includes: a bake oven capable of operating at $125 \pm 5/0$ °C, a vacuum chamber capable of operating at 60 mbar, and an acoustic microscope.

This test method is performed after devices have been subjected to reflow simulation and 22 or more devices have been submitted for AM scans as per the flow in B.2.

- a. Bake the devices for 4 hours at 125 °C
- b. Perform AM scan (use for comparison to AM scans performed on same devices after reflow simulation and after water penetration)
- c. Within a vacuum chamber, place the devices onto a plate above a beaker of water
- d. Evacuate the vacuum chamber down to a pressure of 60 mbar
- e. Let samples sit for 15 minutes
- f. While still in the vacuum chamber transfer the devices from the plate to the beaker of water
- g. Let samples sit for 15 minutes
- h. Return the vacuum chamber back to atmospheric pressure
- i. Let samples sit for 15 minutes
- j. Remove the devices from the beaker
- k. Perform AM scan

Any alternate flow is acceptable as long as similar results are obtained. One alternative is to initially place the devices into the beaker of water prior to evacuating the chamber.

Flux application can be omitted from the preconditioning flow only if there is no reduction in the amount of delamination observed between the 2nd and 3rd AM scans for any of the devices subjected to the water penetration test.



Figure D.1 — Acoustic microscopy (AM) scans of samples at different steps in the water penetration test

As the amount of delamination detected decreased for the device on the right in the AM scans shown in Figure D.1, this means that water was able to enter this package, thus a path does exist, and thus flux application would be required.

Annex E (normative) Dye penetration test method

E.1 Purpose

This test method assesses the capability of encapsulated devices to withstand contaminant ingress by detecting external flaws (voids and cracks) in encapsulating materials and substrates as well as separations at material interfaces. This test method, more specifically the technique described in MIL-STD-883 Method 1034, has been shown to repeatedly detect cracks and separations of 1 micron in width with additional documentation suggesting capability down to 0.5 micron or even less. This test method can be used on unstressed devices to assess initial workmanship and package integrity and on stressed devices to assess damage caused by preconditioning and reliability stress testing.

E.1.1 Difference between dye penetration and “Dye and Pull”

Dye penetration testing is different from “Dye and Pull” testing (formally called “Dye and Pry”) which is typically used to assess the integrity of solder joints. For Dye and Pull testing, the solder joints to be assessed are immersed in a very low viscous dye (typically red in color), subjected to a vacuum, and then dried so as to stain any exposed surface within the solder connection. The device is then mechanically pulled off of the board. It is assumed that each solder connection will separate at its weakest location, which should be any location that has an externally exposed defect (void, fatigue crack, weak interface, etc.) or is the weakest mechanical point/interface within the joint. After the device is removed, the red dye should have visibly stained the surface of any externally exposed defect, but all other fracture surfaces due to the “prying” operation should not exhibit any red dye. As this red dye is not mixed with a hardening material, this process will not yield good results with cross-sectioning techniques, as the red dye will smear.

E.2 Apparatus

The apparatus for the Dye Penetrant test will vary depending on the method used; either the device is first impregnated with a low viscous crack filler that contains a fluorescing dye and then encapsulated with potting material, or the device is encapsulated with potting material that contains a fluorescing dye. Most methods will require the use of a vacuum chamber capable of operating at 60 mbar, while others may require a pressure chamber or both. A bake oven capable of operating at 125 +5/-0 °C will also be required for hardening the crack filler and/or potting materials. This test method will require the use of proper lab facilities and equipment for the preparation and handling of the dye, crack filler, and/or potting materials. This test method will also require the use of cross sectioning and polishing equipment, optical and UV microscopes, and possibly a scanning electron microscope, along with digital imaging equipment for documentation.

E.3 Procedures for impregnation of dye into cracks and delaminated interfaces

The following generalized procedures provide basic information for performing each variation of this test method. The exact procedure will vary based on the technique used, dye and materials used, and technique used for evaluating the extent of dye penetration. Variations are acceptable as long as similar results can be obtained.

E.3.1 is a technique that utilizes a low viscosity, anaerobic liquid penetrant (crack filler) which is mixed with a fluorescent powder and an anaerobic accelerator. This mixture enters cracks and delaminated interfaces within a device and is then hardened. The sample is then potted and evaluated.

E.3 Procedures for impregnation of dye into cracks and delaminated interfaces (cont'd)

(MIL-STD-883 Method 1034 is one example of this technique.). E.3.2 is a technique that utilizes a potting material to which a dye is mixed, and the dye and potting material enter cracks and delaminated interfaces within the device and the potting material is then cured and the device is evaluated. For both techniques a vacuum is used to assist the crack filler or potting material to enter cracks and delaminated interfaces.

E.3.1 Low viscous crack filler mixed with dye method

- a. Mix dye, penetrant, and accelerator into a solution
- b. Place devices into a beaker of the dye solution
- c. Place the beaker into a vacuum oven
- d. Subject beaker to appropriate vacuum conditions
- e. Remove the parts from the dye solution after venting vacuum chamber
- f. Wipe off the parts to remove the excess dye solution
- g. Harden the dye solution at the appropriate conditions

If the samples are to be cross-sectioned, they must first be potted which follows this basic flow:

- h. Mix potting resin and curing agent into a solution
- i. Place devices into a mounting cup
- j. Pour solution over specimen
- k. Place mounting cup into a vacuum oven
- l. Subject mounting cup to appropriate vacuum conditions
- m. Place mounting cup into cure oven
- p. Subject mounting cup to appropriate cure conditions

E.3.2 Potting material mixed with dye method

- a. Mix dye, potting resin and curing agent into a solution
- b. Place devices into a mounting cup
- c. Pour solution over specimen
- d. Place mounting cup into a vacuum oven
- e. Subject mounting cup to appropriate vacuum conditions
- f. (optional – subject sample to pressure)
- g. Place mounting cup into cure oven
- h. Subject mounting cup to appropriate cure conditions

Any alternate flow is acceptable as long as similar results are obtained.

NOTE1 Regarding the cleaning of the outside of the samples (per E.3.1 item f), dye may continue to bleed out of delaminated interfaces for a few seconds to a few minutes. It would be beneficial to identify these samples and document the locations of any bleed out, as these would be points of interest where subsequent destructive analysis (E.4.1 or E.4.2) should be focused.

NOTE2 A marker element may also be added to the penetrating material along with the fluorescing dye. Doing this would enable a second method for tracking the ingress of the penetrating material beyond UV light as stated in E.4.1 and E.4.2. Several elemental mapping techniques (e.g., ToFSIMS, AES, XPS) could be used to look for the marker element within the sectioned or decapsulated sample.

E.4 Assessing dye penetration within sample

The depth to which the penetrating dye has reached within the samples can be assessed in two ways.

E.4.1 describes the use of standard cross-sectioning techniques of cutting and polishing the sample. This technique can be used on all package styles. E.4.2 describes a technique of mechanically opening the impregnated package along the horizontal interface of the molding compound and leadframe/substrate. This technique is best suited for packages for which all interfaces are mainly on one plane, such as transfer molded BGAs and molded leadframe based packages in which the leads exist at or near the mid-point of the package sides (mainly SO and QFP package styles) and do not have any features exiting the top or bottom of the package (such as lands, exposed die pads, and heatslugs/heatspreaders).

Chemical (wet etch), laser ablation, and plasma etch decapsulation procedures shall not be used as the dye would be removed along with the encapsulation materials and thus could permanently modify, or outright eliminate the attributes/regions of interest.

E.4.1 Cross-sectioning (cutting/polishing)

Standard cross-sectioning procedures can be performed on the samples, though make sure that stray particles of dye do not smear or become embedded in the cross-sectioned surfaces. This may give a false representation of the dye in the sample. (Smearing looks like little stars when viewed through the ultraviolet filtering.) Be sure to clean the samples thoroughly after grinding and polishing operations to avoid these complications.

View the sections using a microscope and the appropriate light source to best highlight the fluorescing dye.

The location and direction of the sectioning is chosen based on suspected locations of cracks and delaminated interfaces as shown in the initial AM scans with the location verified by performing x-ray on the potted sample. See Annex F for examples.

E.4.2 Mechanical decapsulation (opening, peeling, cracking, etc.)

This decapsulation method can be used in cases where inspection of the inner surfaces of the device's physical construct is pertinent.

The optimum mechanical decapsulation technique is dependent on a myriad of variables pertaining to the package construct, materials used, and the region of interest to be analyzed. No single technique is equally effective in all cases.

Mechanical decapsulation should be performed only by demonstrably proficient personnel. If the analyst's proficiency for a specific package type is not current, or if it the first time a specific package is being mechanically decapsulated, practice runs should be performed on similar (dummy) units until proper efficiency has been attained.

NOTE Mechanical decapsulation, at specific interfaces inside the package construct (along die-surface, leads-surface, die-pads) can be achieved by cutting with a knife (accurate positioning of the cut-height may ask for special tooling). Another option is to grind the package construct from one or more sides to reach a specific area for successful cutting (e.g., QFN from all sides to separate the exposed pad).

E.4.2 Mechanical decapsulation (opening, peeling, cracking, etc.) (cont'd)

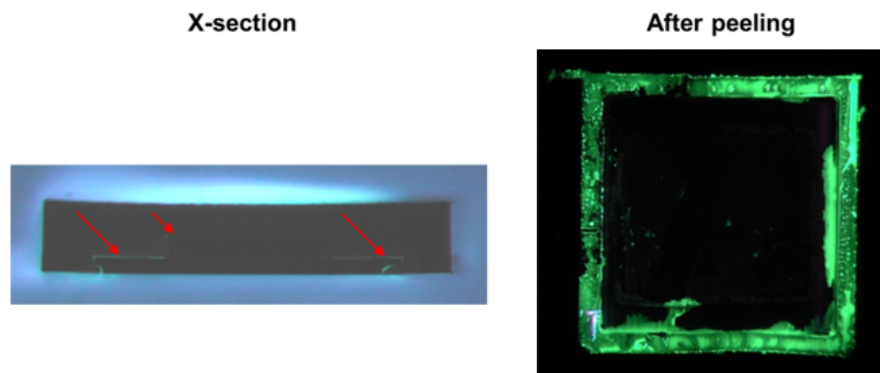


Figure E.1 — Examples of samples submitted for dye penetration after cross sectioning and mechanical decapsulation

E.5 Pass/Fail Criteria

Use the criteria in B.1 to evaluate the depth of the penetrating dye in the sectioned samples. Flux simulation is required if any one sample fails any of the criteria in B.1.

Annex F (informative) Guidance for identifying planes for cross-section

As referenced in B.2 and E.4.1, this annex provides guidance for determining where to perform cross sections on samples that have been subjected to dye penetration and potted to facilitate cross sectioning. It is strongly recommended to perform AM and x-ray scans of the device prior to performing dye penetration, as this will allow for alignment of internal and external structures of the device prior to performing the initial cut of the sample.

Even though the AM scan may not be able to provide information for all interfaces, especially vertical interfaces, it can still provide significant benefit for this analysis by stating whether any delamination is detectable on any horizontal surface. If delamination is detected, one or more cross sections should be performed on the sample to determine if the penetrating dye was able to reach any of the critical features noted by items 1, 2, or 3 in B.1 (the die, a wirebond, a flipchip bump, or any other structure that could degrade and cause the device to not function properly). With that in mind, the construction the device should be reviewed to identify the location of all surface breaking features and internal critical features.

EXAMPLE 1 Device in “SMC” package

As seen in the physical outline drawing for the SMC package in Figure F.1, the two leads that exit on opposite sides of the package are the only surface breaking features. For this device the die is soldered to the lower and upper leadframe, thus the only critical feature would be the die. Therefore, the section should focus on detecting the fluorescing dye on any of the paths that would connect to the die and on the die itself.

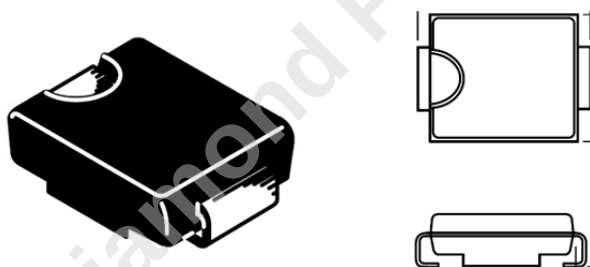


Figure F.1

Figure F.2 and Figure F.3 are AM scans of this package and Figure F.4 is an x-ray image of the package. The AM scan shows delamination on both leads all the way to the die. The green line on the x-ray image shows where the initial cross section was performed on this device as it includes both leads and the die.

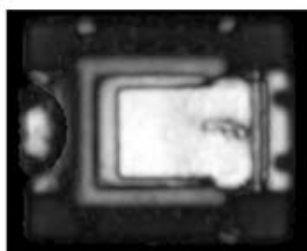


Figure F.2

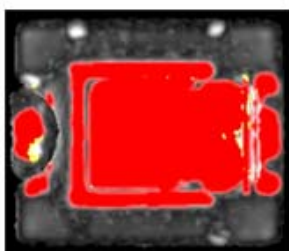


Figure F.3

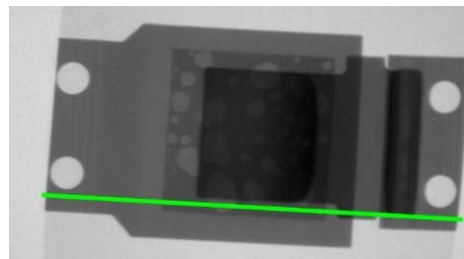


Figure F.4

EXAMPLE 1 Device in “SMC” package (cont’d)

Figure F.5 is a “black and white” image of the cross sectioned device after dye penetration using an UV microscope in which the dye is fluorescing bright white. Note that in Figure F.5 the dye was able to reach the left side of the die and the top of the die, thus one or more external paths existed from the outside of this package to the die (the critical feature within this package).

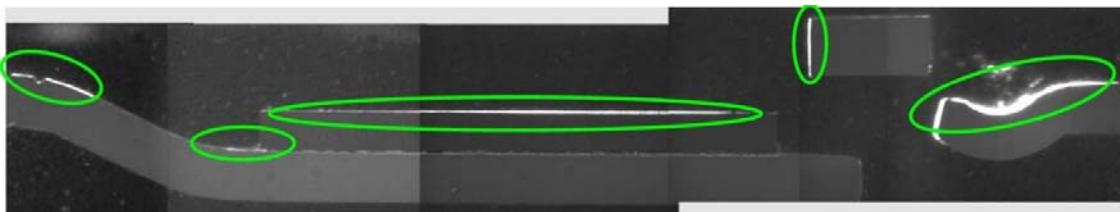


Figure F5

In conclusion, as the dye was able to reach a critical feature (the die), one or more external paths must exist. As this was detected in the first cross-section, no other cross sections were required; and because of the findings, flux application would be required for this device.

EXAMPLE 2 MOSFET device in package with bottom terminations

Figure F.6 and Figure F.7 are side and bottom images of the device which show that the only surface breaking features are the three terminations. For this device the die is soldered to the lower and upper leadframe, thus the only critical feature would be the die. Therefore, the section should focus on detecting the fluorescing dye on any of the paths that would connect to the die and the die itself. Figure F.8 is an AM scan that was performed through the top surface of the package and shows internal delamination in multiple locations within the package.



Figure F.6



Figure F.7

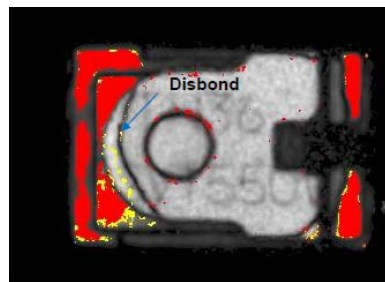


Figure F.8

EXAMPLE 2 MOSFET device in package with bottom terminations (cont'd)

Figure F.9 is an x-ray of the package and when it is compared to the AM scan it appears that there is delamination on the top surface of the die. As was the case for Example 1, the die is soldered to the large pad and to a clip that is connected to the two smaller terminations, thus the only critical feature again would be the die. Therefore, the focus for the cross sections would be to determine if the dye is able to reach any surface of the die.

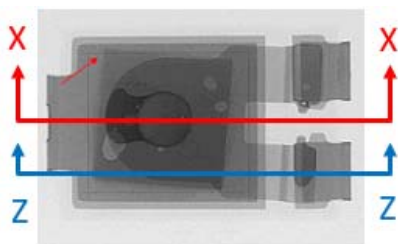


Figure F.9

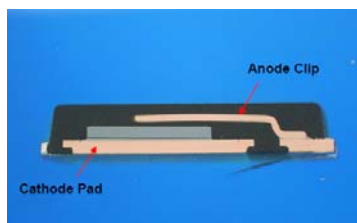


Figure F.10

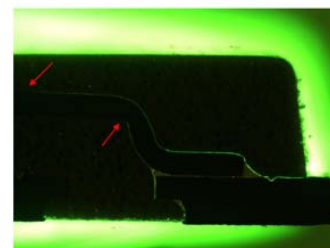


Figure F.11

For this example, two sectional planes were chosen and are shown in Figure F.9 over an x-ray image of the device. The first section (Z-Z), shows the connection of one of the smaller terminations to the anode clip and part of the die and the cathode pad. The second section (X-X) is further into the package and shows the connection of the anode clip to the top of the die. Figure F.10 is an optical image of the first section and Figure F.11 is an image of the same section, but at a higher magnification and using an UV microscope in which the dye fluoresces green. Figure F.11 shows that the dye is able to penetrate along the termination to the anode clip, but due to the location of the section it cannot be shown whether it is able to reach the die. The figure also shows the right edge of the cathode pad, but no dye is detected in the image.

Figure F.12 and Figure F.14 are optical images of the second section (X-X), and Figure F.13 is an image of the same section, but at a higher magnification and using an UV microscope in which the dye fluoresces green. In Figure F.13 the dye was able to reach the left side of the die and bridges from the solder connection of the cathode to the solder connection of the anode, thus shorting this MOSFET die.

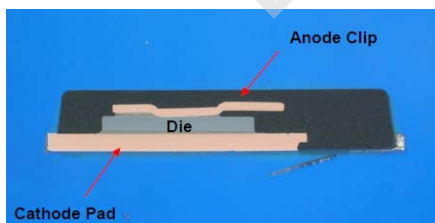


Figure F.12

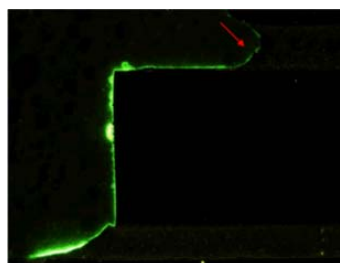


Figure F.13

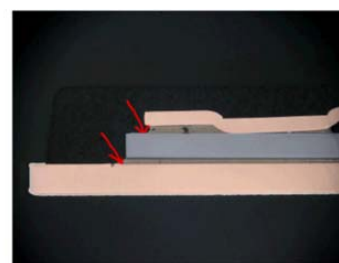


Figure F.14

In conclusion, as the dye was able to reach a critical feature (the die), at least one external path must exist; and because of the findings, flux application would be required for this device.

EXAMPLE 3 TSSOP with bevel and rounded edges

For this TSSOP package that was used in Example 2 of Annex C, the only surface breaking features are the lead fingers and tie bars that exit the sides of the package as shown in the AM scan in Figure F.15 and the x-ray image in Figure F.16. Based on the construction of this device wire bonded stacked die the features of concern would be the two dies and the connection of the wire bonds to the lead fingers.

The AM reflection mode scan image suggests that only the lead fingers may be delaminated, thus there is concern that it may extend to the wire bond on the lead fingers. As the AM image does not show any delamination on the tie bars, one or more cross sections are needed just along the lead fingers to determine if the dye is able to reach a wire bond. The red line on Figure F.16 shows where the initial cross section was performed.

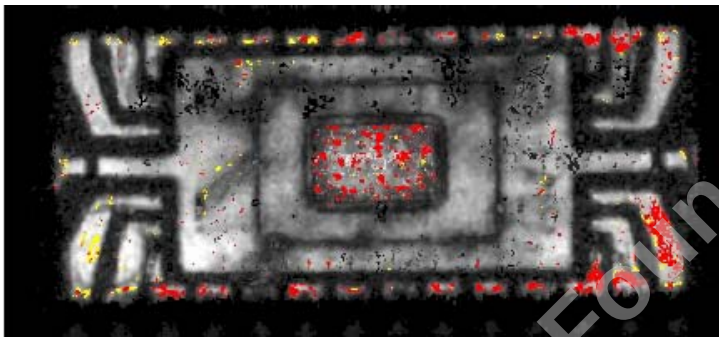


Figure F.15

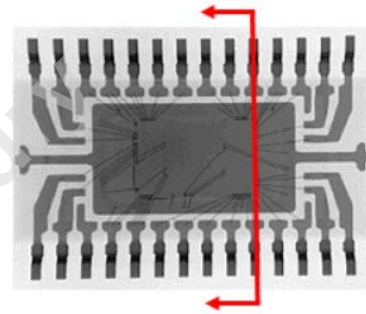


Figure F.16

Figure F.17 is an optical image of the cross-section after dye penetration. The fluorescing images in Figure F.18 show that the entire top surface of the lead finger is delaminated, but the section did not include where the wire is bonded to the lead finger; however, Figure F.19, which is a higher magnification of F.18, shows that the wirebond itself is delaminated as dye is shown around the wire.

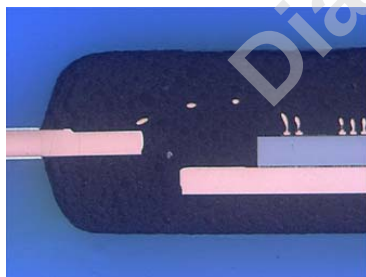


Figure F.17

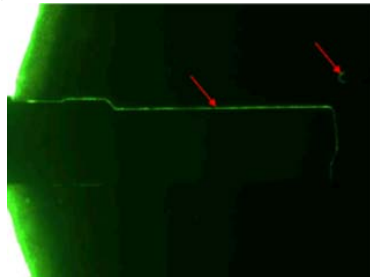


Figure F.18

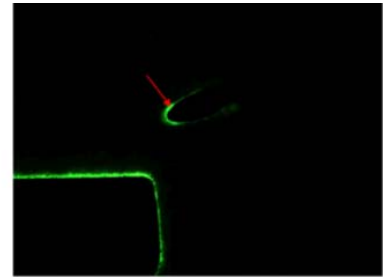


Figure F.19

In conclusion, as the dye was able to reach a critical feature (the wire bond), at least one external path must exist; and because of the findings, flux application would be required for this device.

Annex G (informative) Acoustic Microscopy: edge and surface effects

G.1 How surface topology and defects effect the acoustic microscopy (AM) signal response

The AM technique requires a flat reference surface. Typically, the upper surface of the package or the die surface can be used as a reference. In some packages beveled or curved edges can cause difficulty in analyzing defects near the edges and below their surfaces. For the two images in Figure G.1, the first image shows the path of acoustic signals going to the surface of three differently shaped surfaces, while the second image shows how those acoustic signals are reflected. The green arrows in the second image denote only those signals that would be received by the AM transducer.

**Effects of Surface Roughness
and Sample Geometry**

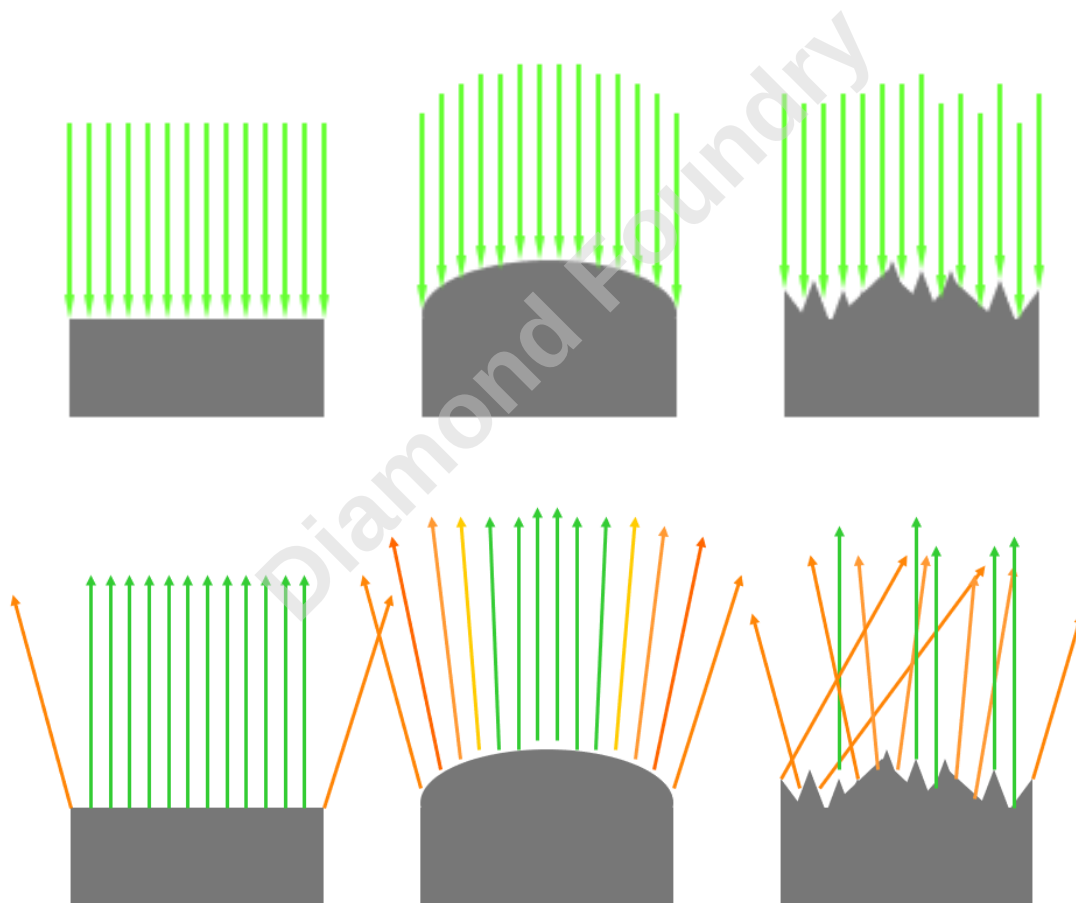


Figure G.1 — Examples of how AM signals respond to different surface topographies

G.2 How (perpendicular) edges effect the acoustic microscopy signal response

The deeper the focus within the package from a vertical edge, the more an edge effect occurs due to the F# (angle) of the focused transducer. The edge effect can be minimized by defocusing (raising the transducer), selecting a higher F# transducer, or inspecting the sample from the opposite side.

The dimension “x” in Figure G.2 shows the area where the edge effect may prevent AM from obtaining an image depending on the depth of interest.

Edge Effect with Depth in Sample

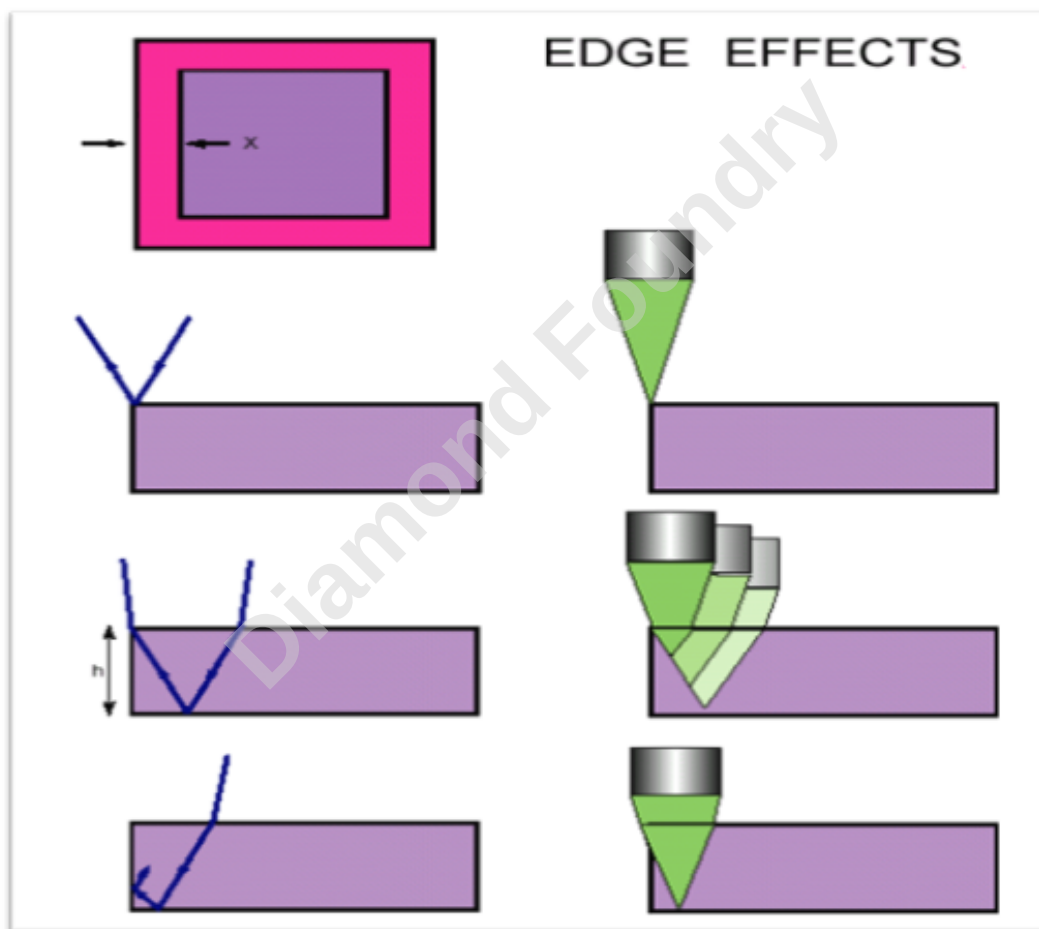


Figure G.2 — Example of how package depth affects the image obtained due to edge effects

G.3 Example of edge effects on TSSOP package

The TSSOP package in this example has stacked die. The AM reflection mode image (Figure G.3) attained suggests that there might be some delamination on the lead fingers, but at initial review, does not appear to be too excessive. Unfortunately, due to edge effects, roughly 2/3 of the lead finger cannot be imaged using AM.

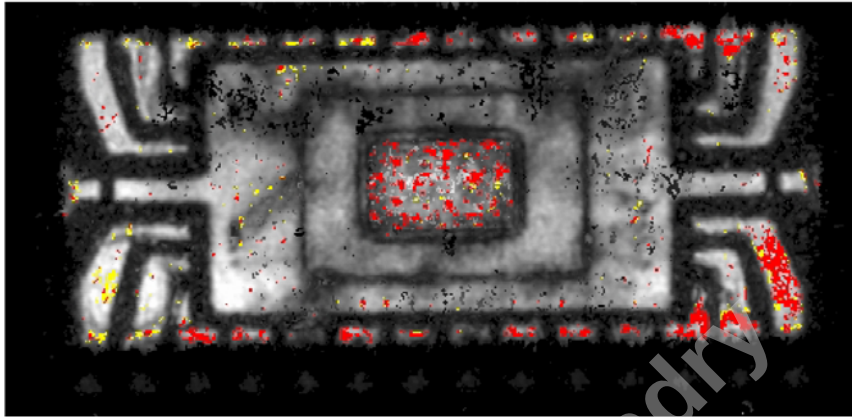


Figure G.3

In the images (Figure G.4), the optical image of one bank of leads of this TSSOP package are aligned with the leads in an x-ray image of the package directly below it. The blue lines that overlay the x-ray and AM images approximate the exterior of the package, based on the top optical image and the x-ray image; and the orange lines approximate how far into the package AM is first able to image the leadframe. The space between the blue and orange lines approximate the area in which the AM signal is not reflected back to the transducer, and thus is not able to generate an image.

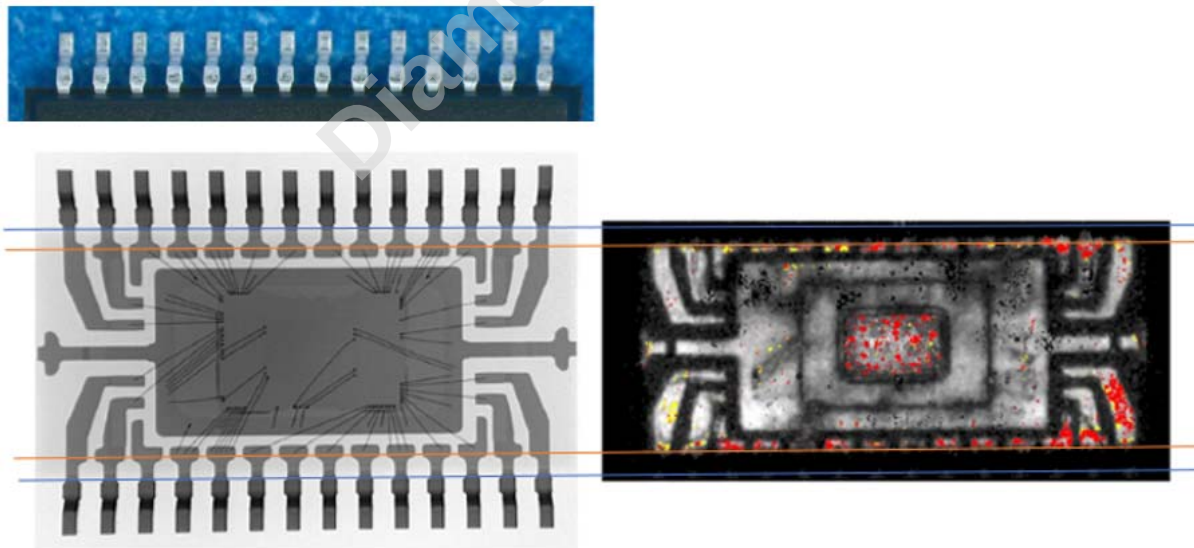


Figure G.4

G.3 Example of edge effects on TSSOP package (cont'd)

The horizontal lines in the left image of Figure G.5 approximate the top and bottom surfaces of the TSSOP package. The vertical line approximates the point where the curvature of the package edge begins. Therefore, any AM signal to the left of the vertical line is not reflected back to the transducer, and thus AM is not able to generate an image. The vertical line in the right image shows the amount of the delaminated internal lead finger (left of line) that is not shown in the AM image due to edge effects.

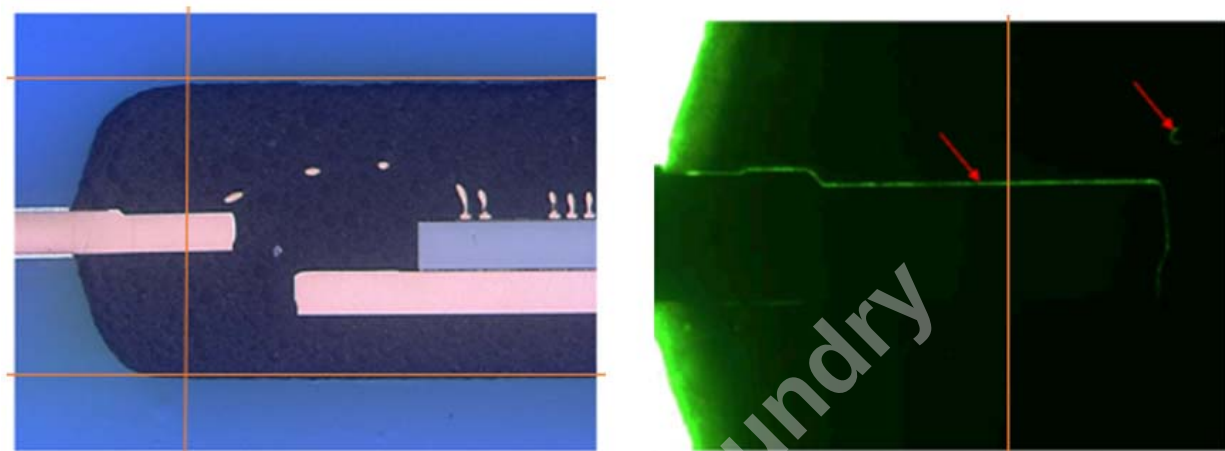


Figure G5

Annex H (informative) Differences between JESD22-A113I and previous revisions

H.1 Differences between JESD22-A113I and JESD22-A113H

The following list briefly describes most of the changes made to entries that appear in this standard, JESD22-A113I, compared to its predecessor, JESD22-A113H (November 2016). If the change to a concept involves any words added or deleted, it is included. Punctuation changes may not be included.

Clause	Description of change
Scope	Added sentence that socketed device are out of scope
2	Updated title of J-STD-075 and added reference for JESD22-B101
3	Inserted a new clause 3, “Definitions”, added definitions for the terms “crack” and “delamination”, renumbered all subsequent clauses, and updated all references to clauses that had been renumbered
4.7	Added new sub-clause under Apparatus for Acoustic Microscope
5.2	Added that inspection be performed “per JESD22-B101 or agreed to alternative”; added that mechanical defects cannot be rejected if no corrective actions are planned to be implemented; and added new paragraph that provides guidance for optional time zero acoustic scans of all samples
5.6.1	Added reference to clause 5.9
5.7	Changed reference for LGAs from a general restriction to flux to stating that only LGA packages intended for soldering do require flux application. Added statement that flux application is only required for high humidity stress tests.
5.7.1	Added this new subclause and moved existing text regarding exemptions to flux application to this sub-clause. Added height restriction for BGA exemption. Slightly modified the text for why flux exemptions are given
5.7.1	Added paragraph that allows option that flux application can be omitted if it can be shown that device has no paths for flux or other contaminants to enter package. Procedure to perform this check is in new Annex B
5.10	New subclause stating when within preconditioning flow that optional post reflow visual and acoustic microscopy inspections are to be performed
7	For item a), added option that reflow profile could be per J-STD-075; for item d), added statement regarding whether flux was required; and added new item h), report differences in criteria and conditions for visual inspection and/or acoustic microscopy scan
Annex A	Updated table to include all changes made to process flow (which are stated above)
Annex B	Added new normative annex, “Process for determining if flux application must be performed”
Annex C	Added new normative annex, Criteria for determining if AM can be used to verify all criteria in B.1”
Annex D	Added new normative annex, “Water penetration test method”
Annex E	Added new normative annex, “Dye penetration test method”
Annex F	Added new informative annex, “Guidance for identifying planes for cross-section”
Annex G	Added new informative annex, “Acoustic Microscopy: edge and surface effects”

H.2 Differences between JESD22-A113H and JESD22-A113G

The following list briefly describes most of the changes made to entries that appear in this standard, JESD22-A113H, compared to its predecessor, JESD22-A113G (October 2015). If the change to a concept involves any words added or deleted, it is included. Punctuation changes may not be included.

Clause	Description of change
4.5	Clarified scope of note to include bumped die; included the condition of when reflow preheat effectively dries out the device; and allowed alternative soak duration based on absorption data
4.6.1	Added cautionary note regarding the potential for inflicting damage during reflow simulation of bumped die
6	Added new item b) that alternate soak conditions/durations be documented



Standard Improvement Form**JEDEC JESD22-A113I**

The purpose of this form is to provide the Technical Committees of JEDEC with input from the industry regarding usage of the subject standard. Individuals or companies are invited to submit comments to JEDEC. All comments will be collected and dispersed to the appropriate committee(s).

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1. I recommend changes to the following:

☐ Requirement, clause number _____

☐ Test method number _____ Clause number _____

The referenced clause number has proven to be:

☐ Unclear ☐ Too Rigid ☐ In Error

☐ Other _____

2. Recommendations for correction:

3. Other suggestions for document improvement:

Submitted by

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